

Risk Assessment and Management of Electricity Markets: A Review with Suggestions

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Abstract—Power industry deregulation has changed the determination of electricity prices from administration approval to market clearing. Affected by multiple random factors, the market prices may fluctuate sharply or significantly increase under extreme conditions, jeopardizing the normal operation of markets. In order to effectively deal with market risks, it is essential to develop the corresponding risk-control mechanisms, especially for spot market pilots. Therefore, this paper presents an overview on the risk assessment and management schemes which are widely used in the electricity markets. First, the key issues for evaluating market risks through inherent uncertainties and market power are summarized, including analysis models and quantitative metrics. Moreover, a comprehensive review of risk management and market supervision in typical electricity markets is presented. Specifically, a multi-time risk-control framework utilized by regulators is introduced to demonstrate the available schemes to reduce market risks. In addition, the various derivative contracts and portfolio optimizations are reviewed to help market participants hedge against market risks. Finally, suggestions for the development of risk management measures in spot market pilots are proposed. The summarized experience in this paper can provide useful references and guidelines for developing risk-control mechanisms in electricity markets.

Index Terms—Electricity markets, market power, risk assessment, risk management, risk sources.

I. INTRODUCTION

SINCE the last decade, the restructuring of the electric power industry has been occurring throughout the world [1]. The major objective of power system deregulation is to introduce competition into the power industry and provide more choices for market participants in the trading of electricity and ancillary services [2]. Due to its advantages in stimulating efficiency and promoting competition, electric power system deregulation originated in the United Kingdom (UK) and further expanded to other countries, such as South America, Australia and Scandinavia [3]. In China, the new

round of power industry reform was launched in 2015, with the publication of “Several Opinions on Further Deepening the Reform of Electric Power System” (No. 9 Document) [4]. Under the guidance of No. 9 Document, different categories of reform practice have been implemented, including the calculation of transmission and distribution prices, the construction of long-term contract markets, etc. [5]. In the next stage, the development of electric spot markets is in progress through the selection of eight provinces and regions as pilot areas, indicating that a competitive electric market will occur in the future in China [6].

Despite the benefits of efficiency improvement for power system deregulation, the introduction of electric markets can also bring about many risks in power trading. Being different from the fixed electricity price set by administrators in conventional systems, the market price varies with time, which is determined by the relationship between supply and demand [7]. Generally, the volatility of electricity market prices is within an anticipated range, considering the regular load fluctuation. Nevertheless, the market prices can fluctuate sharply or significantly increase under extreme conditions caused by different random factors. The risk sources inducing extreme market prices can be divided into three categories: market design flaws, inherent uncertainties, and participant behaviors [8]. Specifically, the market design flaws primarily refer to the unreasonable market mechanism that does not agree with power industry realities [9]. The inherent uncertainties represent the stochastic behaviors of electric elements in power system operations, such as the rapid increase of electric demands [10], device failures [11] and the volatility of renewable energy sources [12]. The participant behaviors refer to the bidding activities of generation units for achieving profits [13].

A failure to deal with the different types of risk sources may result in inestimable social and economic losses, as shown in Fig. 1. It was reported that the widespread electric blackouts in California resulted from multiple risk sources [14]. First, the total electricity consumption in California grew rapidly with an annual increase rate of 4 percent from 1998, driven by economic development [15]. Additionally, no power plants were built from 1989 to 1999 and power from neighboring states cannot be imported due to constrained transmission capacity, leading to the shortage of generating capacity in California [14]. Further compounding this problem, power generators exercised market power by strategically withholding some capacity from the market [16]. The increased de-

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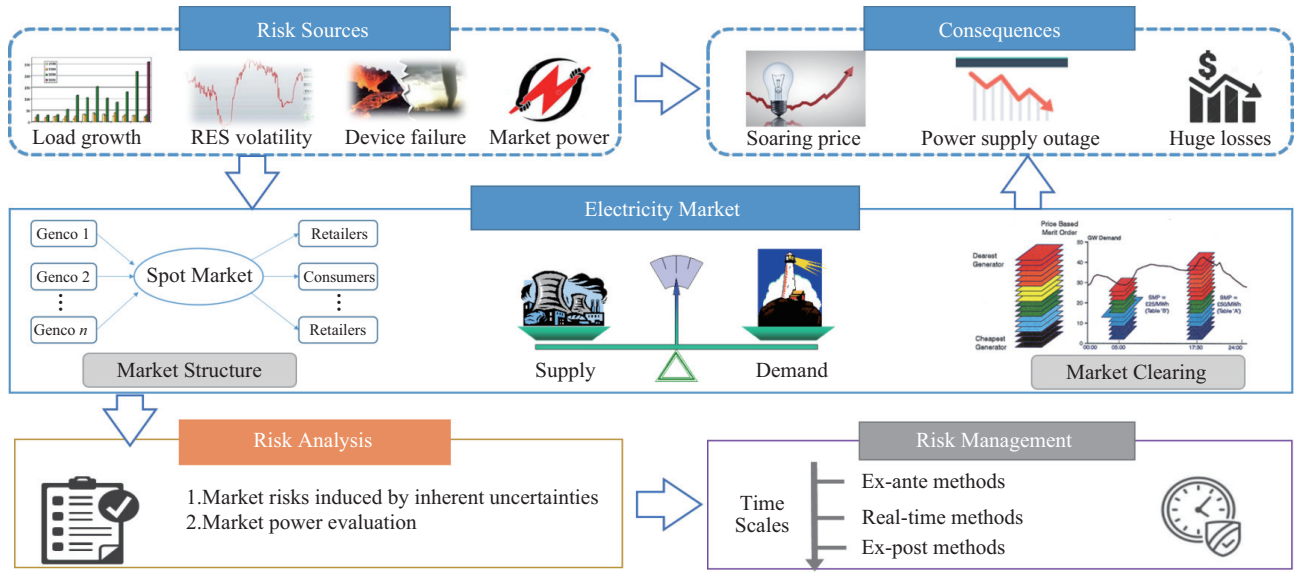


Fig. 1. The structure of this paper.

mand, insufficient power supply and the exercise of market power certainly contributed to the sharp increase in market price. Meanwhile, the California Public Utilities Commission nullified the long-term contracts and encouraged market participants to exchange electricity from the spot market for enhancing market liquidity [17]. Due to market design flaws, the electric companies were forced to buy electricity based on expensive spot prices and sell it for low retail rates. Under these circumstances, the debt of electric companies significantly increased at a pace of as much as \$50 million per day [14]. Finally, several electric companies, e.g., Pacific Gas & Electric, had no choice but to file for bankruptcy protection.

The poignant example of California blackouts entails the urgent need to effectively cope with multiple risks in market operations. Therefore, it is imperative to make a comprehensive review on the risk assessment approaches and management measures. The summarized experience can not only help the existing electricity markets improve relevant risk management rules, but also provide effective references for the development of risk prevention mechanisms in pilot areas. Reference [18] made a survey on the risk management techniques that are widely used in electricity markets, including hedging contracts and portfolio optimization. In reference [19], various risk-hedge financial instruments in PJM and Nord Pool were reviewed to illustrate the price risk management measures. Reference [20] reviewed the utilization of different electricity financial instruments to mitigate market risks for generators and retailers. The previous studies primarily focused on summarizing the risk management measures and illustrating the utilization of different risk-hedge instruments. However, the risk evaluation of electricity markets with respect to different categories of risk sources were not considered. Moreover, the summarized approaches are primarily applicable to market participants to hedge the market price risk. However, the market regulators and operators are more focused on how to develop schemes to reduce market risks, which is seldom summed up in the previous studies.

This paper presents a comprehensive review on the risk assessment and management methodologies that are widely used in typical regions. Considering the diversity of risk sources, the techniques to evaluate the impacts of inherent uncertainties and participant behaviors on market risks are first reviewed. Moreover, the risk management schemes for market regulators and participants are reviewed. On the one hand, the multi-time risk-control framework is introduced to demonstrate the schemes utilized by regulators to reduce market risks, including ex-ante, real-time and ex-post methods. On the other hand, various risk hedging measures utilized by market participants are summarized, including signing derivative contracts and trading portfolio optimization. Based on the summarized experience, various suggestions for the development of risk management schemes, in the pilot areas, are proposed.

II. RISK ANALYSIS IN ELECTRICITY MARKETS

In order to effectively deal with multiple risk sources, one primary task is to evaluate the impacts of random factors on the market and system operational risks. As illustrated in the introduction, the risk sources can be broadly classified as inherent uncertainties and participant behaviors. In this section, the techniques to evaluate market risks caused by inherent uncertainties are first summarized. Regarding participant behaviors, the strategic bidding activities of market participants are acceptable if clearing prices fluctuate within reasonable ranges. System operators are more focused on the specific bidding that profitably sustain prices above marginal costs, i.e., market power of market participants. Therefore, the evaluation models and monitoring measures of market power are then reviewed in this section.

A. Analysis of Market Risks Induced by Inherent Uncertainties

Considering the diversity of inherent uncertainties, the electric market risks induced by various factors have been analyzed

in previous studies. First, the impacts of market structure on market risks are evaluated, including bilateral contract market [21], Poolco market [22], [23], and hybrid market models [24]. It can be concluded that electric consumers could better manage price risks since they had more choices to buy electricity either from a centralized power pool or directly through bilateral contracts. In addition, market risks caused by random failures of electric devices are analyzed, such as generator outages [11] and transmission line failures [25], [26]. According to the analysis results, the device failures can significantly raise the expected price and increase the probability of extreme prices. Moreover, reference [27] analyzes the double effects of device failures and elastic demands on the price risks of electric markets. It can be concluded that elastic demands can stabilize market prices to prevent extreme prices. Furthermore, the price risks of electric markets with high penetration of renewable energy are assessed in [28]–[30]. The analysis results show that the introduction of renewable energy can decrease the expected market price, while increasing the fluctuation of the clearing price.

In addition to investigating the relationship between inherent uncertainties and market risks, the impacts of allocating different resources on price risks have been analyzed. The impacts of reserve arrangements on market risks are evaluated in [31]. It can be concluded that arranging more reserves can significantly decrease the probability of market price spikes. Reference [32] introduces the demand-side resources to reduce the volatility and uncertainty of market prices.

In order to evaluate market risks induced by multiple random factors, two critical issues need to be addressed. The first is the selection of evaluation models, while the second is the development of metrics to quantify the market risks. In the following sections, the analysis models and quantitative metrics of market risk evaluation are summarized.

1) Analysis Models of Market Risks

Different methodologies have been proposed for risk evaluation of electric markets, including simulation techniques, analytical approaches and predicting methods [8]. The analytical methods usually evaluate market risk indices by direct mathematical solutions. In the previous studies, the state selection technique [11], [22], [28], the reliability network equivalent method [30] and the universal generating function [33] are three typical analytical techniques that have been used to analyze market risks. Taking the state selection technique as an example, the market risk R can be evaluated according to the probability p_m and clearing results LS_m of each contingency state m . The system state probability Pr_m can be calculated directly considering the variation component states, e.g. component failure or wind power fluctuation. Consider the impacts of component failures, the Pr_m can be determined as $Pr_m = \prod_{s=1}^F p_s \cdot \prod_{s=F+1}^{N_s} (1 - p_s)$, where p_s is the unavailability of component s . F represents the number of failed components. The clearing results LS_m for contingency state m can be evaluated using the market clearing model. On the basis, the market risk R can be expressed as:

$$R = \sum_{m=1}^{N_m} Pr_m \cdot LS_m \quad (1)$$

where N_m represents the number of system contingency states.

The simulation techniques estimate the risk indices by simulating the market stochastic process behaviors, which are based on the random numbers generated by the computers [34]. The Monte-Carlo simulation (MCS) technique [25] has been widely used in market risk analysis by sampling the initial conditions of each variable. The system contingency state m can be sampled according to the state probability of each component. Based on the sampled state m , the clearing results LS_m can be evaluated using the market clearing model. For M sampled states, the market risk can be evaluated as:

$$R = \sum_{m=1}^M LS_m / M \quad (2)$$

The predicting methods are more focused on describing the relationship between inputs and outputs without exploring the underlying processes. Based on the historical data of electricity prices and demands as inputs, the predicting methods learn the complex relationships between prices and impacting factors, and further forecast the distribution of future market prices. Given the input price dataset denoted as $X = \{x_1, \dots, x_T\}$ the nonlinear map function $f(\cdot)$ is used to predict the output price distribution.

$$y = f(X) \quad (3)$$

To characterize the map function $f(\cdot)$, numerous methods have been proposed, such as the neural network method, machine learning technique, etc. In [35], the neural network method is utilized to predict the locational marginal price in PJM systems. Reference [36] forecasts the distribution of volatility utilizing the extreme learning machine. The detailed illustrations about the recent progress of electricity price forecasting methods can be referred to in the review articles [37] and [38].

Based on the previous discussions, it can be concluded that the simulation techniques and analytical approaches usually evaluate market risks by establishing mathematical models with system operating constraints and solving them under different contingency states [34]. In contrast, the predicting methods analyze market risks based on historical data, where the operational characteristics of power systems are seldom considered [8]. Therefore, it can be noted that more detailed data concerning network structure and system operations are required for simulation techniques and analytical approaches. The predicting methods may be more suitable for the cases without acquiring sufficient system operational data.

2) Quantitative Metrics of Market Risks

The market risks are usually characterized by the volatility of clearing prices and the occurrence of extreme prices. In order to quantify the impacts of random factors on market operations, different metrics have been proposed in previous studies. The expected price and standard deviation of market clearing prices are utilized in [11], [22], [28], [31] to evaluate the price risks. The expected price $\bar{p}$ is a weighted average of clearing prices for different conditions, which can be

represented as:

$$\bar{p} = \sum_{j=1}^{N_s} p_j \rho_j \quad (4)$$

where p_j and ρ_j denotes the probability and electric price of contingency state j , respectively.

The values of standard deviation σ can evaluate the extent of price fluctuating around its expected value, which can be calculated as:

$$\sigma = \sqrt{\sum_{j=1}^{N_s} (\rho_j - \bar{p})^2 p_j} \quad (5)$$

The probability of price spikes defined in [24] is to evaluate the possible extreme prices in the electric market.

$$PPS = \sum_{j=1}^{N_s} p_j \mathbf{1}(\rho_j \geq \rho^{\text{set}}) \quad (6)$$

where $\mathbf{1}(\text{True}) \equiv 1$ and $\mathbf{1}(\text{False}) \equiv 0$. ρ^{set} is a preset threshold of the price spike.

The value at risk (VAR) defined in [39], [40] is to estimate the expected losses due to market movements in a particular period for a given probability of occurrence, which can be represented by the following equation.

$$\text{Prob}(\Delta L \leq VaR) = \alpha \quad (7)$$

where VaR is the maximum expected loss for market participant. $\text{Prob}(\Delta L \leq VaR)$ is the probability that the losses ΔL are smaller than VaR . α is the confidence level.

In reconstructed power systems, market participants are more focused on the nodal price risks than system-wide risks. Due to limited transmission capacity and the uneven distributions of generation sources and demands, the clearing price can differ at different nodes. Hence, the aforementioned price risk metrics are defined on nodal levels in [11], [22], [24], [31]. Moreover, the customers have choices on prices and risks in electric markets. Specifically, some customers may prepare to pay more to receive lower risks while others may be willing to pay less for higher risks. Therefore, the nodal metrics can better characterize the spatial difference of price risks compared to system-wide indices.

B. Analysis of Market Power for Participants

Market power can be defined as the ability of market participants to profitably drive prices over competitive levels for a significant period of time [41]. Due to enormous profits caused by increasing prices, generators are motivated to exercise market power by changing their offer curves. In real operations, the exercise of market power usually includes economic withholding and physical withholding [42]. The economic withholding denotes that a generator submits an offer curve with relatively high prices above its marginal cost with the expectation of gaining higher profits [43]. The generators engage in physical withholding when they reduce the offers of generation capacity to make some proportion of their plants unavailable [42]. In order to ensure the competitiveness and

fairness of electric markets, it is essential to effectively detect and prevent the exercise of market power.

Two categories of analysis methodologies are proposed to evaluate market power, including market concentration indices and simulation methods [13]. The first method calculates the indices directly representing the market concentration, while the second method conducts simulation or experiments to estimate market-clearing prices.

1) Calculating Market Concentration Indices

A great deal of market concentration indices has been proposed to measure market power. The typical indices, such as Herfindahl–Hirschman Index (HHI), Lerner Index (LI), Residual Supply Index (RSI) and Must-Run Ratio (MRR), have been widely used in market power analysis [13]. The HHI is defined as the sum of the squares of market shares of all participants [13], which can be formulated as:

$$HHI = \sum_{g=1}^{N_g} S_g^2 \quad (8)$$

where S_g is the market share of generator g in percentage and N_g is the number of generators. This index is relatively simple since it cannot reveal the impacts of demand variation and transmission constraints.

The LI is utilized to measure the deviation of price from the firm's marginal cost [44], which can be measured as:

$$LI_g = (\rho_g - mc_g) / \rho_g = 1 / \varepsilon_g \quad (9)$$

where ρ_g and mc_g are market price and marginal costs of generator g at the profit-maximizing output, respectively. ε_g is the elasticity of demand seen by generator g . Compared to HHI, the LI can reflect the effects of demand elasticity on market power; however, it cannot take the impacts of transmission constraints into consideration.

The RSI measures the percentage of remaining generation capacity in the market after subtracting the capacity of one generator from the total values [45].

$$RSI_g = (TP - P_g^{\max}) / D \quad (10)$$

where TP is the total supply capacity and D is the total demand. $P_g^{\max}$ represents the capacity of generator g . Similarly, the impacts of transmission constraints cannot be considered in the RSI.

To consider the line constraints, MRR is proposed in [41] by calculating the capacity of each generator that must be provided to supply given loads in a congestion zone.

$$MRR_{hg} = \frac{D_h - P_{hl} - \left(\sum_{g=1}^{N_{hg}} P_{hg}^{\max} - P_{hg}^{\max} \right)}{\sum_{g=1}^{N_{hg}} P_{hg}^{\max}} \quad (11)$$

where P_{hl} is the import limit of zone h . $P_{hg}^{\max}$ is the capacity of generator g in zone h and N_{hg} is the number of generators, respectively. D_h is the total demand of zone h . However, the MRR cannot reveal the controllability of each generator on clearing price considering the impacts of random failures.

To overcome the limits of the previous typical indices, more complementary indices are proposed. In [46], the dynamic HHI ($DHHI_h$) in zone h is defined as the sum of the squares of generation shares in a percentage for all load areas at a certain hour t .

$$DHHI_{ht} = \sum_{g=1}^{N_{gh}} \left(\frac{P_{hgt}^{\max}}{\sum_{g=1}^{N_{gh}} P_{hgt}^{\max}} \right)^2 \quad (12)$$

where $P_{hgt}^{\max}$ is the available capacity of generator g in zone h at hour t . Compared to HHI, DHHI considers the transmission congestions and demand variation.

Effective Lerner index (ELI) is proposed in [47] to consider the impacts of network congestion on the traditional LI.

$$ELI_t = \frac{\text{mean}(\rho_{it}) - mc_t}{\text{mean}(\rho_{it})} \quad (13)$$

where $\text{mean}(\rho_{it})$ represents the mean of all clearing prices at node i and hour t ; mc_t is market participants' marginal cost without considering transmission limits.

Must-run Share (MRS) is proposed in [41] to characterize the minimum market share required for each generator to supply a given load. Before defining MRS, the Must Run Generation (MRG) of generator g representing the minimum capacity of generator g that must be provided to supply system load considering generation and transmission constraints. The MRG P_g^{must} can be calculated by solving the optimization problem in (14).

$$\begin{aligned} \min & P_g^{\text{must}} \\ \text{s.t. } & e^t (P_g - D) = 0 \\ & 0 \leq P_g \leq P_g^{\max} - \\ & Pl^{\max} \leq F(P_g - D) \leq Pl^{\max} \end{aligned} \quad (14)$$

where e is a vector with all ones. P_g and D are the generation output vector and demand vector, respectively. $P_g^{\max}$ and $Pl^{\max}$ are the maximum generation output vector and line limit vector, respectively. F is the distribution factor matrix.

After determining the MRG, the MRS of generator [41] can be defined as:

$$MRS_g = \frac{P_g^{\text{must}}}{D} \quad (15)$$

where D is the total demand in the power market. Compared to MRR, the MRS is determined by solving the optimal power flow model under different contingency states, which can characterize the impacts of load variations and random failures on market power.

2) Experimental and Simulation Methods

The analysis of market power using either simulations or experiments depends on the market-clearing software tools that can simulate the power transactions in electric markets. In [48], a market simulator is developed to simulate the functioning of a pool-based market. Reference [49] proposes a detailed simulator that can model the multi-period market clearing process considering the ramping rates of the generators.

In order to assess market power, experimental methods usually require a few experimenters to model the behaviors of market participants. In [50], undergraduate business students from Cornell University were invited to submit offers and bids in electric markets, based on which the market power was analyzed. The results obtained by experimental methods may be unreasonable since they are highly dependent on the professional qualities of the experimenters. Therefore, simulation methods based on intelligent learning algorithms are developed. An agent-based simulation technique is proposed in [51] to analyze the possibility of market power abuse. In [52], a reinforcement learning model is introduced to simulate the market power of market participants in the day-ahead market.

III. RISK MANAGEMENT AND MARKET SUPERVISION

In order to effectively control market risks, the primary task is to develop reasonable market designs and management mechanisms to reduce price risks. The market operators and regulators are requested to monitor the behaviors of market participants to prevent market power abuse. At the same time, generators and consumers are encouraged to utilize diversified methods to hedge the price risks. Hence, market risk management measures are primarily summarized from two aspects, including control-schemes to reduce market risks and methods of hedging market risks.

A. Control-schemes to Reduce Market Risks

Considering the temporal differences of random factors, the control-schemes to reduce market risks should be developed at different time scales. In this paper, a multi-time risk-control framework is introduced to demonstrate the schemes for risk reduction in the electricity markets, as shown in Fig. 2. Moreover, the risk-control practices in different markets are compared in Table I, including PJM, UK and Australia.

1) Long-term and Medium-term Schemes

Faced with multiple random factors such as demand growth, the widely-used schemes include improving market rules and increasing system capacity. The first measure is to revise the deficiencies of market rules, such as regulating the price caps. The second scheme is to implement generation planning to ensure the adequacy of generating capacity for demand growth.

The price caps are set to restrict the total revenues that companies can charge for services. Generally, both the market clearing prices and the prices offered by generators are restricted by price caps. Practical experience has shown that the price caps can effectively reduce price volatility and sharp price spikes [53]. Nevertheless, the set of price caps that may make the clearing price cannot reflect the true supply-demand of commodities [53]. Therefore, PJM and Australia introduced price caps, whereas the UK does not consider the same measures. It can be noted that the price caps used in Australia are larger than those in PJM. This is primarily because the Australian market wants to help generators recycle their costs to stimulate investments in power generation. Furthermore, the

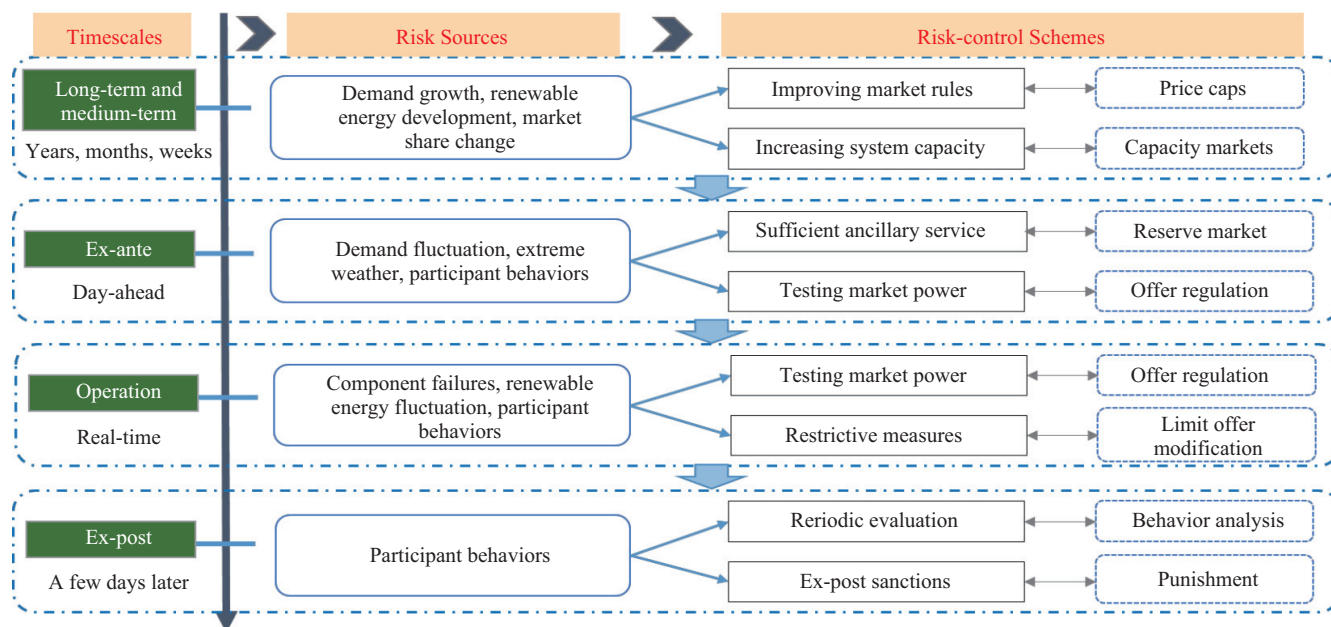


Fig. 2. The multi-time framework to reduce market risks.

TABLE I
THE COMPARISON OF RISK-CONTROL MEASURES IN DIFFERENT ELECTRICITY MARKETS

Time-scales	Practice	PJM	Australia	UK
Long-term and medium-term	Increasing system capacity	Capacity market	PASA	Capacity market
Ex-ante	Price caps	1000 \$/MWh	12500 \$/MWh	/
	Sufficient ancillary service	Reserve market and regulation market	Spinning Reserve response market	Reserve market and frequency response market
Real-time	Testing market power	Calculating market concentration indices (HHI, TPS)	Monitoring the prices offered by generators and physical withholding	Structural screening, conduct-and-impact screening
	Limiting offer modification	Constant each day	The offer capacity can vary at any time	/
	Testing market power	Calculating market concentration indices (HHI, TPS)	Monitoring the prices offered by generators and physical withholding	Structural screening, conduct-and-impact screening
Ex-post	Market investigations and punishments	Monitoring the inappropriate conducts utilizing ex-ante measures	Monitoring specific market events or unusual market behavior	Monitoring by Ofgem through antitrust law, market power license

typical markets adjust the price caps periodically to create a balance between risks and efficiency.

Sufficient capacity is essential to guarantee the long-term stability of electricity prices. Table I shows the typical measures in different markets, including capacity markets and Projected Assessment of System Adequacy (PASA). Based on the market experience in PJM, the UK introduced capacity markets in 2014 to ensure the security of the electricity supply [54].

The capacity markets of the UK take the form of an annual auction for the capacity to be delivered in the upcoming four years. The Australian market performs PASA annually to evaluate the capacity requirements considering multiple random factors, such as demand growth [55]. The capacity analysis information will be published on the web site to provide incentives for market participants. Due to the high price caps, many firms are motivated to make investments for building generation units.

2) Ex-ante Schemes

The ex-ante or day-ahead schemes used to reduce market risks can include ensuring the adequacy of ancillary service and testing market power. The ancillary service markets are all introduced in the typical regions to allocate reserve and frequency response. Nevertheless, the clearing models of ancillary service markets can differ in the three typical regions. Unlike the sequential clearing of ancillary service markets and energy markets in the UK, the reserve and regulation markets in PJM are integrated with the energy markets. Taking the 30-minute reserve as an example, PJM clears the 30-minute reserve market via a simultaneous optimization with the day-ahead energy market [56]. The objective function of the optimization model is to minimize the total costs of energy and reserve subjects to the operating constraints of power systems.

In order to prevent market power abuse, different monitoring methods are utilized in the typical electric markets. In the PJM, market concentration indices are calculated to determine

whether market power exists in a given constraint, such as HHI and the three pivotal suppliers (TPS) test [57]. The TPS test is to determine whether the power needed to relieve the constraints can be met after removing the supply of the test supplier company and the other two largest suppliers. If the TPS test is not passed, the price offered by the corresponding supplier will be automatically set as 110 percent of operating costs. It should be noted that conduct-and-impact screening is also a typical measure used in other markets of the USA [57].

The authority in Australia monitors the exercise of market power by analyzing the reasonableness of prices offered by generators. For example, the prices of the supply curve offered by generators should reflect their reasonable expectation of their short-run marginal costs [55]. The unreasonable prices of the supply curve will be set by the default ones. Moreover, the prices of ancillary services are also monitored according to the corresponding costs. In the UK, structural screening and conduct-and-impact screening are utilized to monitor market power [58]. In accordance with Australia, structural screening is used to compare the offer prices of generators with their marginal prices and the average of historical prices. The conduct-and-impact screening is used to compare the offers of generators with a pre-defined reference level that approximates competitive offers. If such offers exceed the predefined thresholds over those reference levels, the supplier fails to pass the conduct test [58]. Similarly, the offer prices that fail the two tests will be replaced with the reference ones.

3) Real-time Schemes

The widely-used risk-control schemes in the real-time operation of markets can include market power testing and limitations of offer modification. It should be noted that the testing measures of market power in real-time operations are similar with ex-ante schemes. Moreover, the limitations of offer modifications for generators are closely related to the market structure of the three regions.

The spot market in the PJM consists of the day-ahead market and real-time market performed by system operators [59]. Based on generation offers and demand bids, the day-ahead market is cleared to determine the hourly prices considering system operating constraints. The real-time market is cleared to calculate real-time prices based on the actual system operating conditions. Hence, the limitations of offer modifications primarily refer to the differences of offers submitted in day-ahead and real-time markets. In the PJM, the generators selected in the day-ahead market are not allowed to alter their offers in the real-time markets [59]. Only the generators that are available but not selected in the day-ahead market can modify offers.

The spot market in the UK consists of the day-ahead market and balancing market. The day-ahead market is a contract market performed by market operators without considering system operating constraints. The bids and offers submitted by market participants in the day-ahead market have no effects on system scheduling [60]. At a certain time before the balancing market, the generators and demands are required to submit the final physical notification (FPN) of contracted electricity to system operators [60]. Moreover, market participants should

simultaneously submit the bids and offers indicating the willingness to operate a level below and above FPN, respectively. Hence, there are no limitations on the differences of offers between day-ahead and balancing markets.

Being different from PJM and UK, the spot market in Australia only consists of the real-time balancing market. The initial offers of generators are submitted for the next day before 12:30 am. Before the actual dispatch of balancing markets, generators are allowed to change or shift their offer capacity at any time [61]. However, the offer prices are prohibited to be modified. In addition, the generators should submit explanations and reasons for modifying offers, which will be recorded for future investigations [61].

4) Ex-post Schemes

After the operation of markets has passed, the regulators will periodically analyze the conduct of market participants and the associated market results. Once the market participant behaviors are found to violate market rules, the offending participants will be fined or disqualified to participate in future market events. The ex-post analysis results can also provide useful information for regulators about the potential market power. In the UK, the market power license will be given to the participants if the licensee is found to have a position of substantial market power in determining the wholesale prices of electricity [58].

The measures to investigate the market conditions can differ in different regions. For example, the ex-post market investigation in the PJM includes daily, monthly, quarterly and annual analyses of results to identify the exercise of market power and inappropriate conduct [57]. The ex-post analysis measures include the TPS test and conduct-and impact screens utilized in the ex-ante market power investigation. Based on the analysis results, the market rules and ex-ante analysis processes will be adjusted to address the identified conduct.

B. Methods of Hedging Market Risks

Although different control-schemes can be utilized to reduce the price volatility, the market risks cannot be completely eliminated. Therefore, market participants usually sign different types of derivative contracts to hedge market risks. The utilization of contracts for generators can also reduce their motivations to exercise market power, further reducing market price risks [62]. Faced with different trading approaches, portfolio optimization is therefore proposed to optimize the trading strategies to minimize price risks.

1) Signing Derivative Contracts

Different derivative contracts have been introduced by market participants to hedge price risks. The most commonly used contracts include contracts for differences (CFDs), electricity futures, electricity options and financial transmission rights (FTRs) [18]. The CFDs are bilateral agreements between generators and consumers to exchange power P_c with a pre-specified price ρ_c [63]. The difference between contracted power P_c and actual power P_r is settled using spot market prices ρ_r , which can be represented as:

$$FR = \rho_c \cdot P_c + \rho_r \cdot (P_r - P_c) \quad (16)$$

where FR denotes the actual payment of electricity consumers. It should be noted that no physical commodity is transferred between buyers and sellers during the settlements [18].

The electricity futures are standardized forward contracts with a specified trading amount, settlement procedures, delivery time and locations, etc. [20]. Electricity futures have the same payoff structure in (16) as CFDs. In most cases, electricity futures are settled by financial payments rather than physical delivery [20]. The electricity options are contracts that provide the holder the right to buy or sell the commodities at a specified price over a specified period [64]. Nevertheless, the holders of the option are not obligated to trade electricity at the specified time.

The previous contracts are usually utilized to hedge the temporal price risks, whereas the risks of volatile prices at different nodes cannot be avoided. The FTRs are therefore developed to help market participants hedge against locational price risks due to network congestion [65]. The FTRs are financial entitlements that entitle the holder to receive compensation for congestion costs incurred on a specified transmission path defined by its source and sink. The congestion costs FR_{io} can be calculated by the difference of nodal price between its source and sink, which can be represented as:

$$FR_{io} = Q(P_i - P_o) \quad (17)$$

where P_o and P_i represent the nodal prices of source node o and sink node i ; Q represents the contracted volume of FTR. It should be noted that the FTRs do not represent a right for the physical delivery of power through the specified transmission path. With the combination of FTRs and CFDs, the contract holders can simultaneously hedge against temporal and locational price risks.

2) Trading Portfolio Optimization

Portfolio optimization refers to optimally allocating the trading approaches with the goal of making a balance between the desired profits and the corresponding risks. The Modern Portfolio Theory (MPT) proposed by Markowitz is one of the widely-used techniques in portfolio optimization [18]. The technique is to minimize the price risks measured by the variances of the portfolio subject to a given return.

The portfolio of the assets denotes the combination of all potential assets. Let n denotes the potential asset and ω_n denotes the investment proportions of asset n . Supposing that the return rate r_n of each asset n has the probability distribution with expected return $E(r_n)$ and variance σ_n^2 . The expected return $E(r_c)$ of the portfolio can be expressed as $E(r_c) = \sum_{n=1}^N \omega_n \cdot E(r_n)$, where N represents the number of potential assets. Similarly, the variance of the portfolio $\sigma^2(r_c)$ can be expressed as:

$$\begin{aligned} \sigma^2(r_c) &= \sum_{n=1}^N \sum_{m=1}^N \omega_n \omega_m \sigma_{nm} \\ &= \sum_{n=1}^N \omega_n^2 \sigma_n^2 + \sum_{n \neq m} \sum_{m=1}^N \omega_n \omega_m \sigma_{nm} \end{aligned} \quad (18)$$

where σ_{nm} represents the covariance between the returns of the n th asset and the m th asset. On this basis, the portfolio

optimization model can be formulated as:

$$\begin{aligned} \max_{\omega_n} U &= E(r_c) - \frac{1}{2} A \cdot \sigma^2(r_c) \\ \text{s.t. } \sum_{n=1}^N \omega_n &= 1 \\ \omega_n &\geq 0 \end{aligned} \quad (19)$$

where U is the utility function and A denotes the weighting factor representing the risk appetite of decision makers. The proposed technique has been widely used in the risk management of electricity markets. In [66], a risk model based on the mean-variance method is proposed to minimize the short-term price risks of portfolios subject to transaction cost limits, generation unit constraints, and financial constraints. Reference [67] applies the MPT to optimize the energy allocation between the spot market and contract market. In [68], a midterm power portfolio optimization problem with risk management is proposed for load-serving entities to serve their loads. Reference [69] proposes a portfolio optimization model to provide the best investment profile for market players in multiple electricity markets. The previous studies prove that the MPT can be an effective technique for market participants to allocate their investment portfolios.

IV. SUGGESTIONS FOR RISK MANAGEMENT IN PILOT AREAS

The summarized risk analysis and management methods can provide useful references for pilot areas to develop risk-control mechanisms. Moreover, the current conditions of the power industry should also be considered in the development of risk management measures. In this paper, some suggestions for risk management in the pilot areas are proposed. In order to make the proposed suggestions more convincing, the market rules in different pilot areas of China are illustrated

A. Developing a Comprehensive Risk-control Framework

Based on the construction experience in typical markets, the comprehensive risk-control framework should be developed in pilot areas considering different time scales. The designed framework should consist of long-term, ex-ante, real-time and ex-post schemes. The risk-control schemes at different time scales should be designed as a whole since they are closely coordinated with each other. According to the published market rules in different pilot areas of China [70]–[72], the specific schemes and their relationships with power industry realities are illustrated, as shown in Table II.

In long and medium terms, the price caps should be introduced, including the upper limits of offering price for generators and market-clearing price. At the early stage of spot market construction, the price caps should not be set as large values to effectively control the price risks. For example, the upper limits of offering price and clearing price in Zhejiang province are respectively set as 0.8 RMB/KWh and 1.2 RMB/KWh, which refer to the current feed-in tariffs of generators [70]. Being different from the PJM and UK that introduce a capacity market, it is suggested that a system

TABLE II
THE COMPARISON OF RISK-CONTROL MEASURES IN ZHEJIANG, GUANGDONG AND SHANDONG PROVINCES

Time-scales	Practice	Zhejiang	Guangdong	Shandong
Long-term and medium-term	Increasing system capacity	Government approvals	Government approvals	Government approvals
Ex-ante	Price caps	Caps of offering price and clearing price	Caps of offering price and clearing price	Caps of offering price and clearing price
	Sufficient ancillary service	Reserve market and frequency market	Frequency market and reserve market	Frequency market
	Testing market power	RSI test and conduct-and-impact screening	Comparing offered price and reference price	Comparing offered price and reference price
	Market intervention and termination	Clarification of intervening conditions and measures	Clarification of intervening and terminating conditions	Illustration of intervening and control measures
Real-time	Limiting offer modification	Constant each day	Constant each day	Constant each day
Ex-post	Market investigations and punishments	Monitoring and evaluating the inappropriate conducts	Monitoring unusual market behaviors	Monitoring unusual market behaviors

capacity expansion plan continues to be developed by the government at the early state of the spot market. This is primarily because the introduction of capacity markets may make the market rules more complicated.

In the day-ahead market, arranging sufficient ancillary service and testing market power are essential measures to reduce market risks. At the early stage of market construction, the ancillary service is suggested to be purchased through reserve markets and frequency markets. The clearing models of ancillary service can be determined according to the power industry realities of the pilot areas. For example, the frequency market and energy market in Shandong province are cleared separately to simplify the market rules [72]. In Zhejiang province, the ancillary service and energy markets are jointly optimized to improve market efficiency [70]. Regarding market power test and prevention, different measures have been developed in the three pilot areas, as shown in Table II. Taking Zhejiang as an example, there exists one generation company named as Zhejiang Energy Group that accounts for 50% of market share [6]. Considering this, the generating units that Zhejiang Energy Group owns are required to submit their offers separately. Moreover, the RSI test and conduct-and-impact screening are combined to monitor market power [70].

Besides market measures, market intervention and termination are emphasized in the three pilot areas [70]–[72]. In order to respond to the situations in which the electric market cannot deal, the boundary conditions of market termination are defined. Taking Shandong as an example, three categories of market termination scenarios are demonstrated, including the tension between power supply and demand, the malfunction of market operations, and the occurrence of extreme events that the market cannot work out [72]. Furthermore, the response actions of market termination are demonstrated. Specifically, the system operators are responsible for the formulation of a dispatching plan based on system operational conditions, instead of market-clearing results.

In the real-time market, it is suggested that the modifications of generation offers should be restricted, as evidenced by the market rules in three pilot areas. The specific restriction measures of offer modification can be constituted according to power industry realities. For example, the generators in Zhejiang are forbidden to alter their offering prices and capacity and are only allowed to alter their maximum outputs in the

real-time market [70]. After the operation of markets, periodical market evaluation and analysis are essential to monitor and punish inappropriate behaviors of market participants.

B. Gradually Introducing Various Financial Derivatives

In order to help market participants hedge the price risks, financial derivatives should be introduced in electricity markets. The introduction of financial derivatives can also reduce the motivation of generators to exercise market power. In accordance with the development of spot markets, the construction of power finance markets should be steadily promoted step by step.

At the early stage of the spot market, the pilot areas are suggested to adopt necessary financial contracts to simplify market rules considering the lack of experience on market operations. In Zhejiang, Guangdong and Shandong, only CFDs are introduced to help market participants hedge price risks [6]. Moreover, the specifications about the categories, signs and settlements of contracts should be developed according to the power industry realities of pilot areas. For example, two categories of CFDs are developed in Zhejiang, including bilateral contracts and vesting contracts [70]. Bilateral contracts are signed freely by market participants, while the specifications of vesting contracts are administrated by the Zhejiang government. The vesting contracts are converted from the traditional government-regulated generation plans to facilitate the smooth progression of the market reform. In order to reduce the settlement risks of contracts, the signed CFDs are required to be submitted to the exchange center to carry out settlements on their behalf.

With the surge in spot market construction, more varieties of financial contracts can be introduced, including electricity futures, options and FTRs. Taking Zhejiang as an example, the electricity futures and FTRs are planned to be introduced after the stable operation of the early-stage market [70]. In the goal market, electricity options and other financial derivatives will be introduced. In addition, third parties, such as financial organizations, are encouraged to participate in financial markets. The enrichment of contract categories and the expansion of market scales can provide more effective risk management tools for market participants.

V. CONCLUSION

The stochastic fluctuation and extreme values of market prices in spot markets necessitate the need to develop risk management mechanisms. In order to summarize the reference of risk management, this paper first reviews the existing techniques to evaluate the market risks induced by inherent uncertainties and market power. Specifically, the methodologies to analyze the price risks, considering inherent uncertainties, are demonstrated, including simulation techniques, analytical approaches and predicting methods. Regarding analyzing the existence and extent of market power, two categories of analysis methodologies are proposed, including calculating market concentration indices and simulation methods.

Faced with multiply risk sources, a comprehensive review concerning risk management is then presented. On the one hand, a multi-time risk-control framework utilized by regulators is introduced to demonstrate the risk reduction schemes, including ex-ante, real-time and ex-post methods. On the other hand, different risk hedging measures for market participants are reviewed, including signing derivative contracts and trading portfolio optimization. The summarized experiences can provide useful references and guidelines for pilot areas to develop risk analysis and management mechanisms.

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